Data?

What Does "Data" Mean?

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What's a Pirate's Favorite Letter?

R is free software (it's a programming language) that scholars use to analyze data (like running complex statistical analyses and visualizing patterns).

RStudio is also free software that makes R a little easier to work with on the front end.

Why R:

- It's FREE
- It's open source
- It's powerful and adaptable
- It's designed for data analysis and visualization
- It's widely used across multiple disciplines

Some Reminders...

Correlation does not imply causation.

Nomothetic causation:

- The causal link is probabilistic, not deterministic.
- An outlier or two doesn't "disprove" or "refute" the causal theory.
- ...a butt load of outliers might, though...
- To say that X_1 causes Y does not stop other things (e.g., X_2) from being causally linked, too.
  - but the relationship between X_1 and X_2 matters here, because we have to look out for confounding and intervening variables

Data vs. Statistics

Data are individual pieces of information.

- the raw facts and figures, without any sort of analysis yet.
- often associated with quantitative methods, but qualitative methods use data, too.
- e.g.: Alice spent $1. Bob spent $3. Chuck spent $5.

A **statistic** is calculated and interpreted using multiple data points.

- results from analyzing data
- provides direction on interpreting raw data
- e.g.: Alice, Bob, and Chuck spent a combined average of $3. Chuck's purchase made up over 50% of the total money spent. Damn, Chuck.

Data, Datum, Dataset, Database

Datum: singular; one individual piece of information.

Data: plural; a collection of pieces of information along a characteristic.

Dataset: a collection of data on multiple characteristics.

Database: a collection of datasets.

var1	var2	var3
1	4	Purple
2	9	Seven
3	43	Eggnog
4	219	Hector
5	86	Smelly

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var1	var2	var3
a	19	Carrot
b	48	Pound
c	13	Ruby
d	6.33	Flute
q	11	Zaire

How to Spot a Bad Statistic

1. Can you see uncertainty? (e.g.: Question averages)
2. Can you see yourself in the data? (e.g.: Get context)
3. How was the data collected? (e.g.: Check the questions)
4. Look at the data yourself!